National University of Computer and Emerging Sciences
Lahore Campus

Professional Practices in IT (CS 4001)

Date: 25th Feburary, 2026

Course Instructor(s):

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Sessional-I Exam

Total Time (Hrs):	**1**
Total Marks:	**36**
Total Questions:	**4**

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Roll No Section Student Signature

Please answer all questions except question2 on answer sheet in the same order as they appear on question paper. Answer question2 on the question paper and attach question paper with answer sheet. Each part/task answer should not be more than 3 lines. Answer of each task/part more than 3 lines will not be considered. Write to the point.

Question1: **Nexus Global**, a massive "legacy" financial services provider that dominates the traditional banking and credit card processing market, is facing stagnation. Their tech stack is aging, and they are losing Gen-Z users to agile "neobanks."

SecureAI is a cutting-edge, mid-sized cybersecurity firm that has developed a revolutionary **"Neural-Fraud" engine**. This proprietary AI can detect fraudulent transactions with 99.9% accuracy in real-time. SecureAI also holds several highly restrictive government contracts and employs a "dream team" of 40 senior data scientists who are considered world leaders in adversarial machine learning. [12+3]

The Move:

Nexus Global announces a **$\$2.4$ billion takeover** of SecureAI. While the market initially sees this as a simple tech upgrade, industry analysts point out that SecureAI also recently launched its own small-scale "Digital Wallet" which was starting to poach Nexus's high-net-worth clients.

Tasks:

1. Identify which of the following reasons apply to this takeover. For **each** reason selected, you must provide a **specific piece of evidence** from the case study above to justify your choice.

Expanding Range of Offerings, Acquiring New Staff, Eliminating a Competitor, Economies of Scale, Expanding Customer Base

2. Choose the **one** reason from the list above that you believe is the **primary driver** for the $\$2.4$ billion price tag (which is $10 times SecureAI's actual revenue). Justify why this reason outweighs the others in terms of long-term strategic value.

Solution:

Takeover Reason	Specific Evidence from Case Study
Expanding Range of Offerings	Integration of the **"Neural-Fraud" engine** into Nexus's legacy stack. It transforms a standard processing service into a "secure, AI-protected" premium service.
Acquiring New Staff	The acquisition of the **"dream team" of 40 senior data scientists**. In the AI field, "acqui-hiring" is often cheaper and faster than recruiting individual world-class talent.
Vertical Integration	Nexus is moving **backwards in the production path**. Fraud detection is a critical "raw material" (security layer) for a financial transaction. By owning it, they control the quality and cost.
Eliminating a Competitor	The SecureAI **"Digital Wallet"** was directly poaching high-net-worth clients from Nexus. Buying the company removes this existential threat to their user base.
Economies of Scale	Nexus is a "global" provider. Spreading the fixed cost of the AI's development over **billions of Nexus transactions** makes the "per-transaction" cost of security negligible.
Expanding Customer Base	Targeting **Gen-Z users** and reclaiming **high-net-worth clients** who prefer the agility and security of modern FinTech solutions.

Task 2: The "Primary Driver" (Critical Analysis)

While multiple reasons apply, the strongest arguments for the **$\$2.4$ billion price tag** (the "Strategic Premium") usually center on two specific areas:

Option A: The Defensive Move (Eliminating a Competitor)

- **Argument:** Nexus is a "stagnant" giant. The loss of high-net-worth clients to SecureAI's digital wallet wasn't just a loss of revenue; it was a signal of the brand's death.
- **Justification:** The high price is a "ransom" to save their core business. The cost of losing their entire premium market far exceeds the $\$2.4$ billion spent to kill the competition.

Option B: Acqui-hiring (Acquiring New Staff)

- **Argument:** In AI, the technology is often secondary to the people who can iterate on it. 40 world-class scientists are an "appreciating asset."
- **Justification:** Nexus cannot build a "Neural-Engine" with their legacy staff. Buying the "brain power" allows them to pivot their entire global strategy, which is worth far more than the target's current annual revenue.

Question2: Three university friends — **Ahsan (software developer), Bilal (UI/UX designer), and Sana (business/marketing specialist)** — have started an IT startup called **"CodeBridge Solutions."**

The startup develops: Custom web applications for local businesses, Mobile apps for startups, A SaaS product (subscription-based appointment booking system for clinics and salons), Maintenance and hosting services. Initially, the founders worked from home and handled only small freelance projects. However, within 10 months the company has grown quickly. The startup now has: 12 employees (developers, QA tester, and support staff), 18 regular business clients, Monthly revenue of approximately **PKR 1.2 million,** Plans to launch their SaaS product internationally next year. They are now planning to: Rent an office, Sign long-term contracts with hospitals and retail chains, Purchase servers and cloud services, Hire additional employees, Apply for bank financing, Possibly bring an investor within 2 years

Recently, some issues have worried the founders: A client's e-commerce website crashed during a sale, and the client threatened legal action for financial loss. They signed a contract worth PKR 8 million with a hospital management system project. The bank asked them to provide an official business registration before approving a loan. An investor showed interest but asked about company shares and ownership percentage. The founders are currently operating informally and have not registered the business legally. They are confused whether to register as: **Partnership Firm** or a **Private Limited Company [2+8]**

1. Based on the case, recommend whether CodeBridge Solutions should register as: **Partnership Firm** or a **Private Limited Company.**

   CodeBridge Solutions should register as a Private Limited Company.

2. Justify your recommendation by referring to **at least FOUR facts from the case** (Legal risk, Large contracts, Bank financing, Future investment.).

1. Legal Risk (Client Lawsuit)

The case states:

a client threatened legal action after a website crash

If the company is a partnership, the partners would have to personally pay damages.
A private limited company protects their personal assets through limited liability.

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2. Large Contracts

The company signed a:

PKR 8 million hospital management project

Large projects involve high financial and legal risk.
A limited company reduces personal financial exposure of the founders.

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3. Bank Financing

The bank requires:

official business registration before approving a loan

Banks prefer limited companies because:

- they are more stable
- legally recognized
- have formal records

Therefore, a private limited company improves chances of obtaining financing.

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4. Future Investor

The investor asked about:

company shares and ownership percentage

A partnership cannot easily issue shares.
A private limited company allows:

- share distribution
- investment
- ownership transfer

This is essential for attracting investors.

Question3: In England and Wales, only licensed members of recognized professional bodies (e.g., Chartered Accountants) are legally permitted to audit the financial accounts of public companies. This is known as reservation of function, where certain professional work is restricted by law to qualified professionals. [2+6]

Should there be a similar reservation of function for IT professionals where only licensed IT professionals should be allowed to develop life-critical systems? Justify your answer accordingly.

Reasons in Favor

1. Public Safety

2. Accountability and Responsibility

3. Quality and Professional Standards

Has this reservation of function for IT professionals happened globally? Justify your answer accordingly.

Your Reason 1: Hard to define “life-critical”

It is difficult to clearly identify whether a system is life-critical or not.

✔ Correct — and this is a serious practical problem.

Unlike a bridge or a surgery, software is embedded everywhere and its impact changes depending on context.

Example:

- A hospital ventilator control system → obviously life-critical.
- A hospital appointment booking system → normally non-critical.
- But if the booking system fails and emergency patients cannot be scheduled → it indirectly affects lives.

Similarly:

- GPS navigation → convenience
- But aviation navigation software → life-critical

So the same type of software can be:
safety-critical in one context and non-critical in another.

This makes regulation complicated.

Your Reason 2: Who will be the authority?

Which authority will decide whether a system is life-critical?

✔ Also correct.

For doctors → Medical Council
For accountants → Chartered accounting bodies
For lawyers → Bar Council

But software engineering has **no single global governing body**.

Even if a country creates one:

- software is developed internationally
- teams work remotely
- open-source contributors exist worldwide

A Pakistani developer could write code used in a German hospital system hosted on US servers. Which country’s license would apply?

So enforcement becomes extremely difficult.

3. Software Changes Continuously

A civil engineer designs a bridge once.

Software is never finished.

It is:

- patched
- updated
- upgraded
- refactored

If a system is updated every week, do we re-approve it every week?
Licensing models from traditional engineering don't fit software well.

4. Team-Based Development

In medicine:
→ one doctor performs a surgery.

In software:
→ hundreds of people contribute.

Who should be licensed?

- programmer?
- tester?
- UI designer?
- DevOps engineer?
- open-source library author?

Responsibility becomes diffused.

5. Global and Open-Source Development

Much critical software depends on open-source libraries.

Example:
A medical device may use a database library written by a volunteer programmer in another country.

You cannot legally require a license from someone who:

- never worked for the company
- never knew the system would be life-critical

6. Innovation and Industry Resistance

The software industry grows because entry barriers are low.

Mandatory licensing would:

- slow startups
- restrict new developers

- reduce innovation

Governments are hesitant to regulate an industry that drives economic growth.